Task-6:- Documentation-Deploy Multiple Frontend Apps on Apache with SSL (Subdomain Based Hosting)

1. Introduction

1.1 Purpose

This document describes the design, implementation, security configuration, and validation of deploying two frontend applications on an Apache Web Server using subdomain-based hosting with SSL encryption on AWS EC2.

The implementation includes:

- **Apache virtual hosting**
- **Subdomain routing**
- **Let's Encrypt SSL automation**
- **AWS EC2 hosting**

The intended audience includes:

- **CTO**
- **Solution Architects**
- **DevOps Engineers**
- **Cloud Engineers**
- **Technical Operations Teams**
- **Security Reviewers**

This document serves as a complete technical reference for secure multi-frontend hosting using Apache and Let's Encrypt.

1.2 Scope

This documentation covers:

- **Ubuntu server configuration**
- **Apache installation and configuration**
- **DNS configuration and validation**
- **Subdomain-based VirtualHost setup**
- **Frontend application deployment**
- **SSL configuration using Let's Encrypt**

- HTTP → HTTPS redirection
- SSL auto-renewal validation
- Security implementation
- Final verification and testing

Out of Scope

- Application code refactoring
- Multi-region disaster recovery
- Advanced auto-scaling
- Full observability stack
- Containerization (Docker/ECS)
- Infrastructure as Code
- Backend/database deployment

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2. Overview of Deployment Project

2.1 Definition

This project implements secure hosting of multiple static frontend applications on a single Ubuntu EC2 server using Apache Virtual Hosts and SSL certificates issued by Let's Encrypt.

Applications Deployed

- Bookshop
- NotesApp

Hosted Subdomains

- bookshop-pritam.pushkar.cloud
- notesapp-pritam.pushkar.cloud

Benefits

- Secure HTTPS communication
- Logical separation using subdomains
- Cost-efficient single server hosting
- Production-style deployment model
- Automatic certificate renewal

2.2 Goals and Objectives

Primary Goals

- Deploy two frontend applications
- Configure subdomain-based hosting
- Enable HTTPS using Let's Encrypt
- Ensure public accessibility

Objectives

- Configure Apache web server
- Deploy Bookshop app
- Deploy NotesApp app
- Configure DNS
- Enable SSL certificates
- Validate HTTPS access
- Verify auto-renewal

2.3 Constraints and Assumptions

Constraints

- Single Ubuntu EC2 instance
- No load balancer
- Manual Apache configuration
- Time-boxed implementation

Assumptions

- Domain already registered
- DNS access available
- Ports 80 and 443 open
- Applications are static builds

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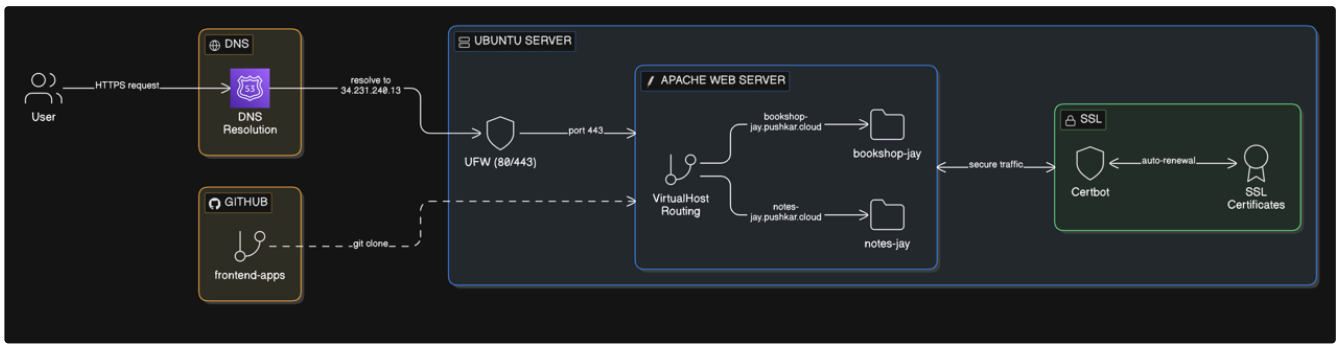
3. Infrastructure and Deployment Architecture



3.1 Infrastructure Overview

Server Type: Ubuntu EC2

Public IP: 18.234.62.119

Web Server: Apache2

SSL Tool: Certbot

Application Directories

```
/var/www/bookshop-vrajlalwala/
/var/www/notesapp-vrajlalwala/
```

3.2 Deployment Flow

Before SSL

User → DNS Resolution → Apache Web Server → Static Files

After SSL

User → HTTPS → Encrypted Traffic → Apache → Application

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4. Pre-Domain Testing (Using Public IP)

Before configuring subdomains, Apache service and application directories were validated.

Verification Performed

- **Apache default page tested**
- **Application directories verified**
- **Service status confirmed**

Result

Apache was running successfully and ready for virtual host configuration.

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5. DNS Configuration and Validation

DNS A records were configured for both applications:

```
bookshop-vrajlalwala.pushkar.cloud → 18.234.62.119
notesapp-vrajlalwala.pushkar.cloud → 18.234.62.119
```

DNS resolution for bookshop subdomain

DNS resolution for notesapp subdomain

Result

Both subdomains resolved correctly to the EC2 public IP.

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6. Apache Virtual Host Configuration

Separate VirtualHost files were created for each subdomain.

Bookshop VirtualHost

```
<VirtualHost *:80>
    ServerName bookshop-vrajlalwala.pushkar.cloud
    DocumentRoot /var/www/bookshop-vrajlalwala
</VirtualHost>
```

NotesApp VirtualHost

```
<VirtualHost *:80>
    ServerName notesapp-vrajlalwala.pushkar.cloud
    DocumentRoot /var/www/notesapp-vrajlalwala
</VirtualHost>
```

Sites were enabled and Apache reloaded successfully.

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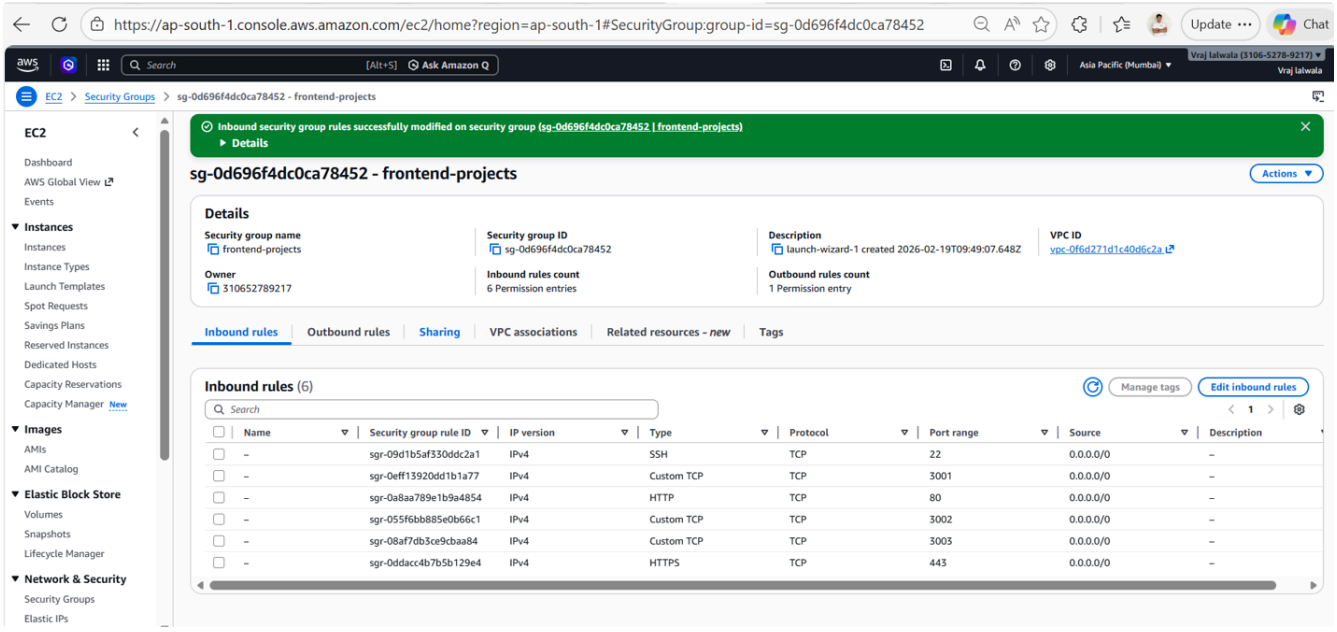
7. Environment & Prerequisites

- **OS:** Ubuntu 22.04 LTS
- **Web Server:** Apache/2.4.x
- **DNS:** A Records pointed to the Server IP for both subdomains.
- **Security:** Ports 80 and 443 opened in Security Groups.



8. Step-by-Step Implementation

Part 1: Apache Installation

1. Updated the package index: `sudo apt update`
2. Installed Apache: `sudo apt install apache2 -y`
3. Verified status: `sudo systemctl status apache2`

```
ubuntu@ip-172-31-34-229:/etc/apache2/sites-available$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-02-23 11:46:01 UTC; 1h 12min ago
       Docs: https://httpd.apache.org/docs/2.4/
    Process: 42790 ExecReload=/usr/sbin/apachectl graceful (code=exited, status=0/SUCCESS)
   Main PID: 40112 (apache2)
      Tasks: 55 (limit: 1008)
     Memory: 12.9M (peak: 14.0M)
        CPU: 1.018s
     CGroup: /system.slice/apache2.service
             ├─40112 /usr/sbin/apache2 -k start
             ├─42795 /usr/sbin/apache2 -k start
             └─42796 /usr/sbin/apache2 -k start

Feb 23 11:46:01 ip-172-31-34-229 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Feb 23 11:46:01 ip-172-31-34-229 systemd[1]: Started apache2.service - The Apache HTTP Server.
Feb 23 12:09:31 ip-172-31-34-229 systemd[1]: Reloading apache2.service - The Apache HTTP Server...
Feb 23 12:09:31 ip-172-31-34-229 systemd[1]: Reloaded apache2.service - The Apache HTTP Server.
Feb 23 12:28:25 ip-172-31-34-229 systemd[1]: Reloading apache2.service - The Apache HTTP Server...
Feb 23 12:28:25 ip-172-31-34-229 systemd[1]: Reloaded apache2.service - The Apache HTTP Server.
Feb 23 12:35:03 ip-172-31-34-229 systemd[1]: Reloading apache2.service - The Apache HTTP Server...
Feb 23 12:35:03 ip-172-31-34-229 systemd[1]: Reloaded apache2.service - The Apache HTTP Server.
ubuntu@ip-172-31-34-229:/etc/apache2/sites-available$
```

Part 2: Repository Cloning & Directory Setup

1. Cloned the repo: `git clone https://github.com/pushkar-iamops/frontend-apps.git`
2. Created directories:
   - `/var/www/bookshop-yourname/`
   - `/var/www/notes-yourname/`
3. Copied files and set permissions:

```
Bash
```

```
sudo chown -R $USER:$USER /var/www/bookshop-yourname/
sudo chmod -R 755 /var/www/bookshop-yourname/

```

Part 3: Virtual Host Configuration

Created two configuration files in `/etc/apache2/sites-available/`.

Example Configuration for Bookshop:

```
Apache
```

```
<VirtualHost *:80>
    ServerName bookshop-vrajlalwala.pushkar.cloud
    DocumentRoot /var/www/bookshop-vrajlalwala
    ErrorLog ${APACHE_LOG_DIR}/error.log
    CustomLog ${APACHE_LOG_DIR}/access.log combined
</VirtualHost>

```

1. Enabled sites: `sudo a2ensite bookshop-vrajlalwala.conf notes-vrajlalwala.conf`
2. Reloaded Apache: `sudo systemctl reload apache2`

Part 4: Let's Encrypt SSL Setup

1. Installed Certbot: `sudo apt install certbot python3-certbot-apache -y`
2. Generated certificates:

```
Bash
```

```
sudo certbot --apache -d bookshop-vrajlalwala.pushkar.cloud -d notes-vrajlalwala.pushkar.cloud
```

3. Selected **Option 2** to redirect all HTTP traffic to HTTPS.

```
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
ubuntu@ip-172-31-34-229:/tmp/frontend-apps/Projects$ sudo certbot --apache -d notes-vrajlalwala.pushkar.cloud
Saving debug log to /var/log/letsencrypt/letsencrypt.log
Requesting a certificate for notes-vrajlalwala.pushkar.cloud

Successfully received certificate.
Certificate is saved at: /etc/letsencrypt/live/notes-vrajlalwala.pushkar.cloud/fullchain.pem
Key is saved at:         /etc/letsencrypt/live/notes-vrajlalwala.pushkar.cloud/privkey.pem
This certificate expires on 2026-05-24.
These files will be updated when the certificate renews.
Certbot has set up a scheduled task to automatically renew this certificate in the background.

Deploying certificate
Successfully deployed certificate for notes-vrajlalwala.pushkar.cloud to /etc/apache2/sites-available/notes-vrajlalwala-le-ssl.conf
Congratulations! You have successfully enabled HTTPS on https://notes-vrajlalwala.pushkar.cloud

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
If you like Certbot, please consider supporting our work by:
 * Donating to ISRG / Let's Encrypt:   https://letsencrypt.org/donate
 * Donating to EFF:                    https://eff.org/donate-le
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
ubuntu@ip-172-31-34-229:/tmp/frontend-apps/Projects$ sudo certbot --apache
```

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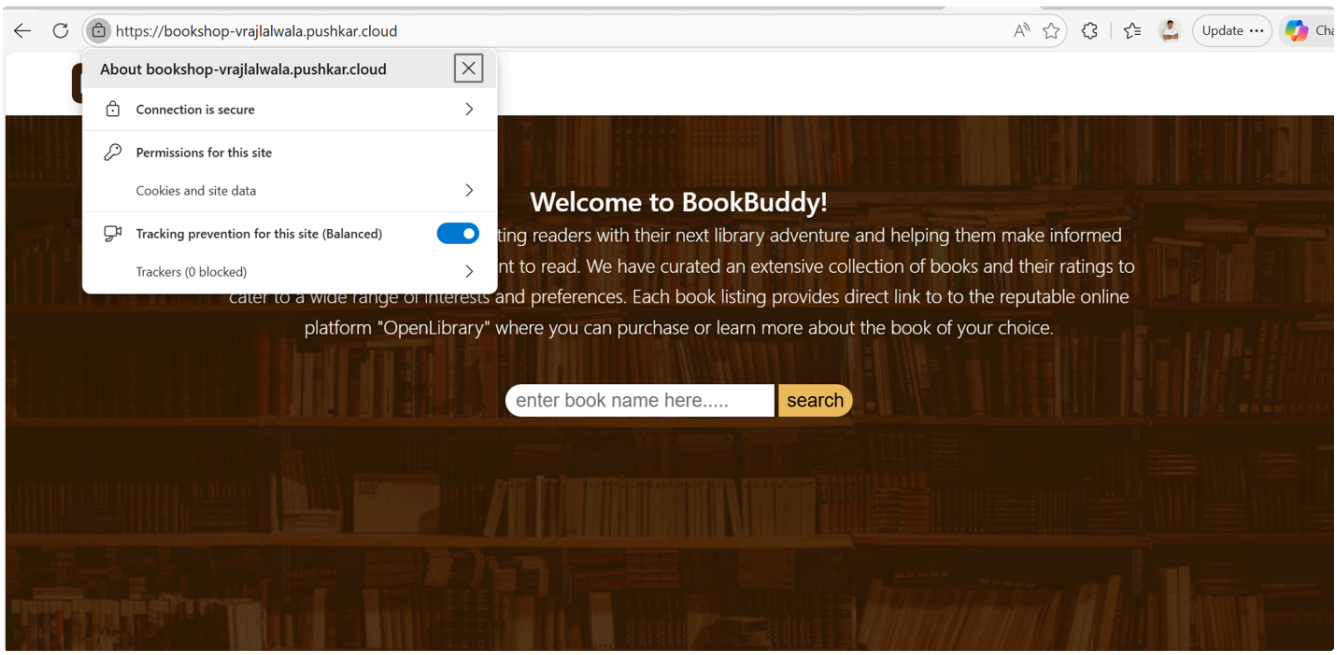
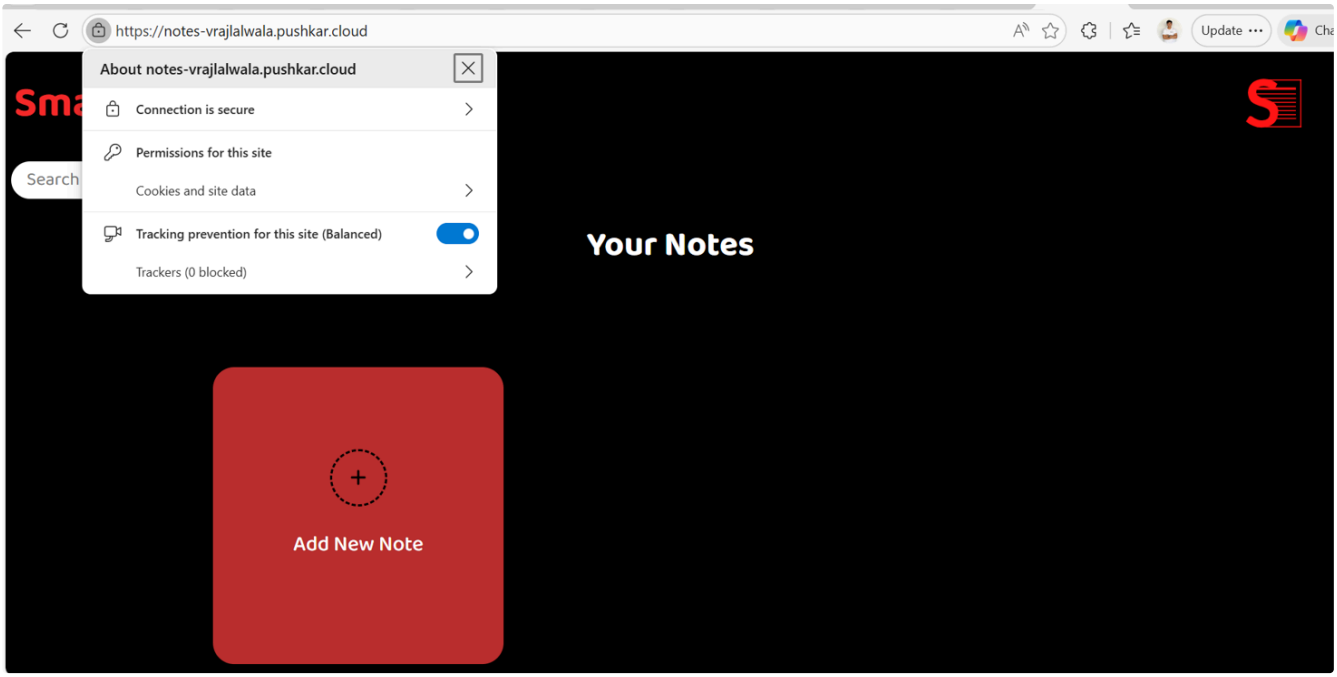
9. Verification & Validation

4.1 SSL Status

The applications are now served over HTTPS.





9.2 Auto-Renewal Test

Verified the renewal cron job/timer:

```
Bash
```

```
sudo certbot renew --dry-run

```

```
ubuntu@ip-172-31-34-229:/etc/apache2/sites-available$ sudo systemctl reload apache2
ubuntu@ip-172-31-34-229:/etc/apache2/sites-available$ curl -I http://bookshop-vrajlalwala.pushkar.cloud
HTTP/1.1 301 Moved Permanently
Date: Mon, 23 Feb 2026 12:39:14 GMT
Server: Apache/2.4.58 (Ubuntu)
Location: https://bookshop-vrajlalwala.pushkar.cloud/
Content-Type: text/html; charset=iso-8859-1

ubuntu@ip-172-31-34-229:/etc/apache2/sites-available$ curl -I http://notes-vrajlalwala.pushkar.cloud
HTTP/1.1 301 Moved Permanently
Date: Mon, 23 Feb 2026 12:40:29 GMT
Server: Apache/2.4.58 (Ubuntu)
Location: https://notes-vrajlalwala.pushkar.cloud/
Content-Type: text/html; charset=iso-8859-1
```

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10. Risk Analysis: Why SSL?

Risk	Mitigation with SSL
Data Sniffing	SSL encrypts data in transit, making it unreadable to attackers.

Phishing/Trust	The Padlock icon verifies site identity to the user.
SEO Ranking	Google penalizes non-HTTPS sites; SSL improves visibility.
MITM Attacks	Prevents attackers from injecting malicious scripts into your static files.

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11. Conclusion

The task was completed within the estimated time. Both apps are isolated via Virtual Hosts and are fully secured.